

## UNIT 4: PRACTICE QUESTION BANK & ANSWER KEY

**Course Pattern:** TE IT - 2024 Pattern (SPPU)  
**Document Type:** Question Bank with Answer Key

**Academic Term:** Semester V (2026)  
**Course Code:** PCC-301-ITT

### Section A: Important Theory Questions

1. **Memory Partitioning & Fragmentation:**

- Differentiate between Internal Fragmentation and External Fragmentation. Explain how Compaction and the Buddy System mitigate fragmentation. [SPPU Nov 2023, Marks: 8]
- Compare First Fit, Best Fit, and Worst Fit dynamic memory placement strategies. [SPPU May 2024, Marks: 6]

2. **Paging, Segmentation, and Address Translation:**

- Explain the hardware address translation mechanism in Paging. Describe the fields of a Page Table Entry (PTE). [SPPU Dec 2022, Marks: 8]
- Explain Two-Level Paging and Inverted Page Tables. How do they reduce page table memory overhead? [SPPU Nov 2023, Marks: 8]
- Differentiate between Paging and Segmentation with respect to block size, user visibility, and hardware table structures. [SPPU May 2023, Marks: 6]

3. **Virtual Memory, TLB, and Thrashing:**

- Explain the step-by-step lifecycle of handling a Page Fault in Demand Paging. [SPPU May 2024, Marks: 8]
- Describe the Translation Lookaside Buffer (TLB) and state the formula for Effective Access Time (EAT). [SPPU Dec 2023, Marks: 6]
- What is Thrashing? Explain how the Working Set Model and Page Fault Frequency (PFF) prevent thrashing. [SPPU Nov 2023, Marks: 8]

### Section B: Numerical Practice Problems

#### Practice Problem 1: Page Replacement Algorithms Trace

Given the page reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

For a system with **3 allocated physical frames** (initially empty), compute total page faults and page hits using:

- (a) FIFO Algorithm [Marks: 4]
- (b) Optimal (OPT) Algorithm [Marks: 4]
- (c) LRU Algorithm [Marks: 4]

#### Practice Problem 2: TLB Effective Access Time Math

A memory system has a RAM access time of 100 ns and a TLB search time of 20 ns. Calculate EAT for:

- (a) TLB Hit Ratio = 85% [Marks: 3]
- (b) TLB Hit Ratio = 95% [Marks: 3]

## Mandatory Final Answer Key

Below are the verified solutions for self-verification:

- **Practice Problem 1 (a) — FIFO Algorithm (3 Frames):**
  - Trace: 1(F), 2(F), 3(F), 4(F), 1(F), 2(F), 5(F), 1(H), 2(H), 3(F), 4(F), 5(F)
  - **Page Faults: 9, Page Hits: 3, Hit Ratio: 25%**
- **Practice Problem 1 (b) — Optimal Algorithm (3 Frames):**
  - Trace: 1(F), 2(F), 3(F), 4(F-rep 3), 1(H), 2(H), 5(F-rep 4), 1(H), 2(H), 3(F-rep 5), 4(F-rep 3), 5(F-rep 4)
  - **Page Faults: 7, Page Hits: 5, Hit Ratio: 41.67%**
- **Practice Problem 1 (c) — LRU Algorithm (3 Frames):**
  - Trace: 1(F), 2(F), 3(F), 4(F-rep 1), 1(F-rep 2), 2(F-rep 3), 5(F-rep 4), 1(H), 2(H), 3(F-rep 5), 4(F-rep 1), 5(F-rep 2)
  - **Page Faults: 10, Page Hits: 2, Hit Ratio: 16.67%**
- **Practice Problem 2 — TLB EAT Math:**
  - (a)  $\alpha = 85\% \Rightarrow \text{EAT} = 0.85 * (20 + 100) + 0.15 * (20 + 200) = 102 + 33 = \mathbf{135 \text{ ns}}$
  - (b)  $\alpha = 95\% \Rightarrow \text{EAT} = 0.95 * (120) + 0.05 * (220) = 114 + 11 = \mathbf{125 \text{ ns}}$

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*End of Operating Systems Unit 4 Practice Question Bank & Answer Key — SPPU TE IT 2024 Pattern*